



GRAPHMD: A TWO-MODULE DIFFUSION FRAMEWORK FOR SMOOTH AND CONSISTENT MOLECULAR DYNAMICS

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Abstract

Molecular dynamics (MD) simulation is a key method for studying complex molecular systems, and deep learning has been widely used to accelerate it. Recently, diffusion models have shown excellent performance in generating high-quality molecular conformations and modeling complex distributions. However, their discrete noise adding process limits the ability to capture smooth oscillatory behavior between consecutive frames and also poses challenges in maintaining spatial structural consistency and effectively processing molecular graph features. To address this, we propose a two-module approach: a molecular graph interaction module, enhanced with classical potential functions, and a diffusion module that uses the Discrete Cosine Transform (DCT) to better capture smooth molecular motions. These improvements enable our model to achieve strong performance across experiments.

Introduction

Molecular dynamics simulates the dynamic behavior of molecular systems based on classical or quantum mechanical principles. Classical MD is computationally efficient but limited in accuracy, whereas Ab Initio MD offers high accuracy at a significant computational cost. Recently, deep learning-based MD has substantially reduced computational expense while maintaining high accuracy. Among various DLMD approaches, denoising diffusion probabilistic models have drawn particular attention for its forward noise-adding process closely corresponds to the introduction of thermal fluctuations and random forces in molecular systems, while the reverse denoising process aligns with energy minimization and structural relaxation on the potential energy surface. This mechanism provides a modeling framework more consistent with the real dynamical processes of molecular trajectories.

Despite the mechanism offers a framework more consistent with the evolution of molecular dynamics and has shown promising applications, it still faces two challenges. First, the discrete noise in existing diffusion models overlooks smooth oscillatory behavior between consecutive frames, which is crucial for understanding conformational transitions and transition states. Second, the adding noise during the diffusion process lacks physical guidance, causing the trajectories to deviate from physically plausible paths.

We propose GraphMD. To address the first issue, we encode temporal trajectories using the DCT to extract dominant low frequency motions. To address the second issue, we further incorporate guiding terms based on Morse and LJ potentials to impose soft physical constraints during graph processing. Before feeding molecular graphs into the diffusion model, we perform feature extraction using a SchNet-like interaction module, which fully leverages interatomic distance information.

Method

Overview

At timestep t , the input molecule is consistently treated as a molecular graph:

$$M^{(t)} = (r^{(t)}, z^{(t)}, u^{(t)})$$

And we follow this fundamental MD simulation procedure, i.e.

$$x^{(t+1)} = f(M^{(t)}, \Delta t)$$

Molecular Graph Interaction Module

The two potential energy terms are obtained from the following formulas:

$$E_{MLJ}^{(t)} = \sum_{i=1}^I V(r) + \sum_{j=1}^J A(r)$$

Gaussian radial basis function (RBF) then fed into the subsequent network:

$$E_{total}^{(t)} = RBF(d^{(t)}) + E_{MLJ}^{(t)}$$

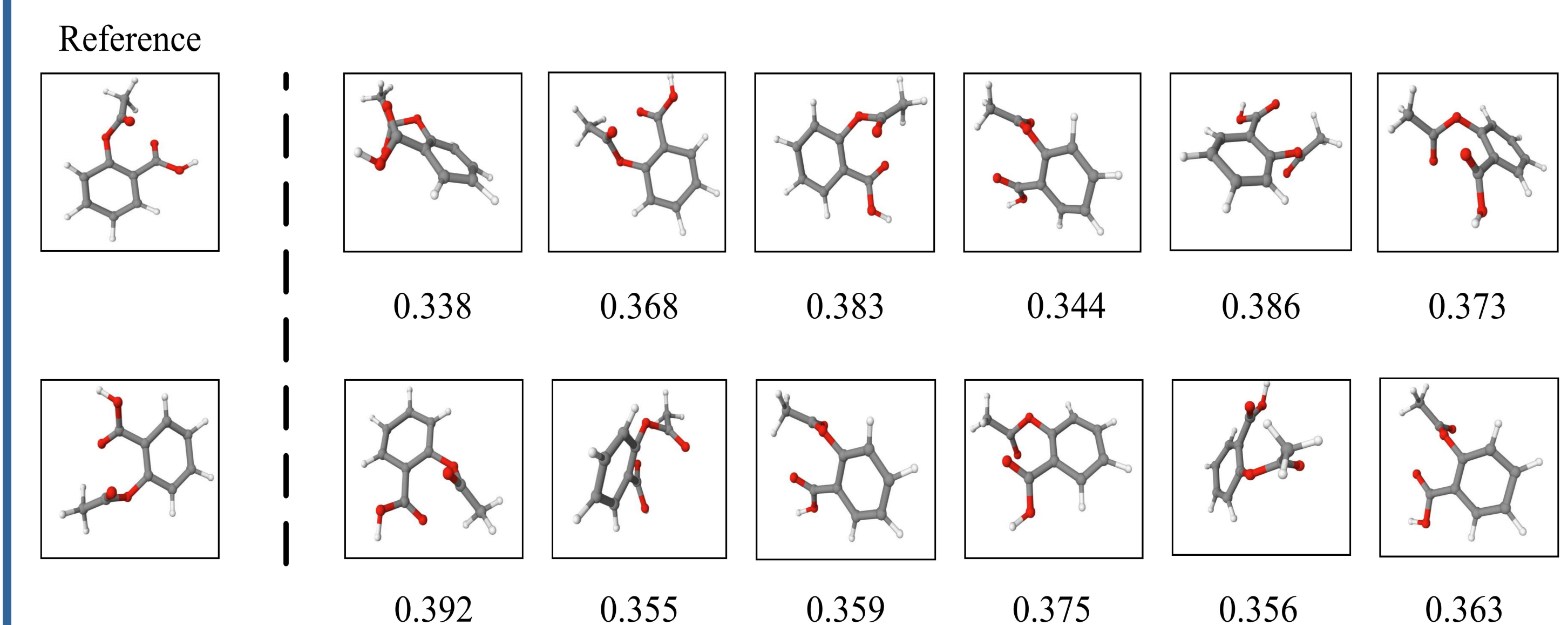
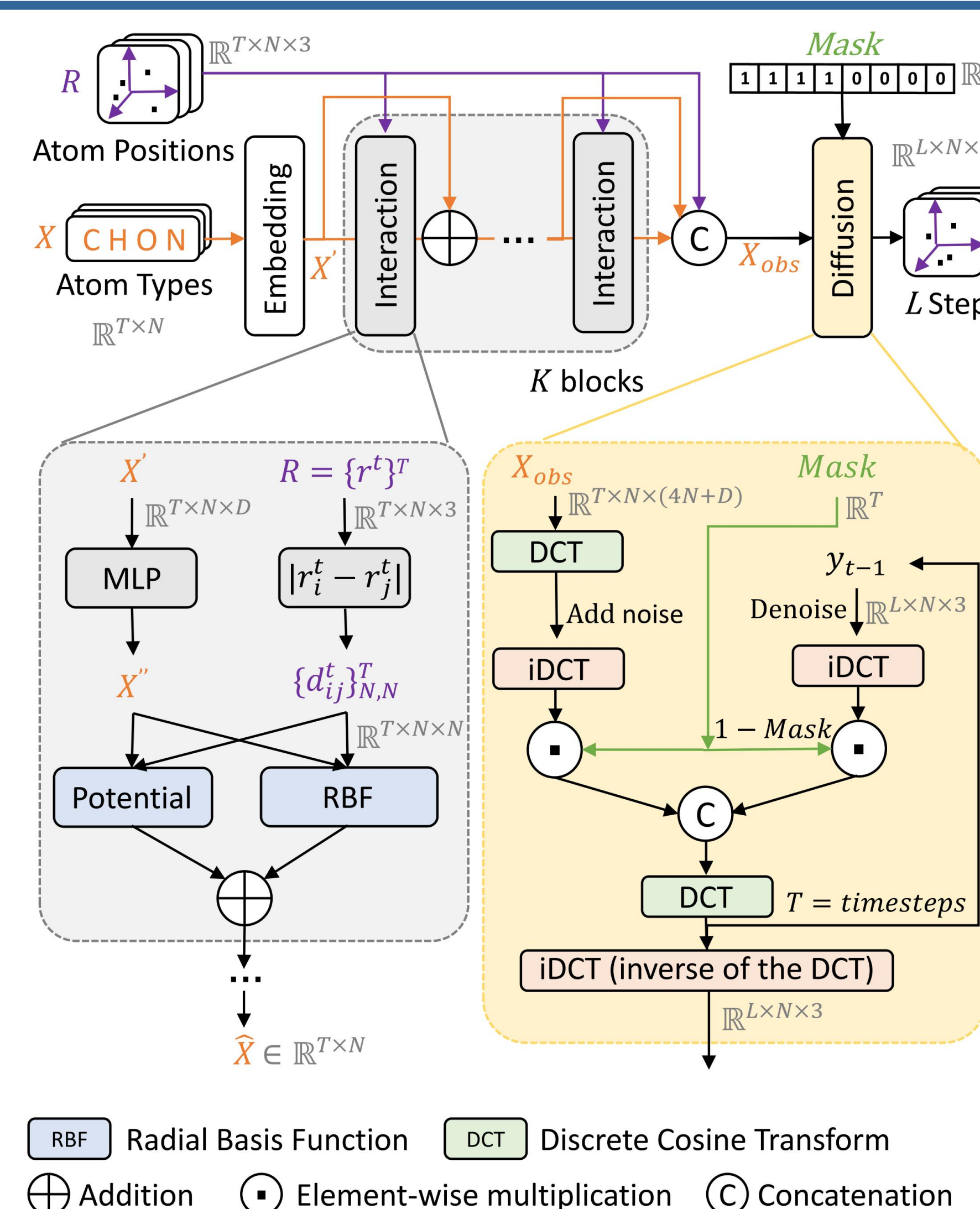
DCT-based Diffusion Molecular Dynamics

We add noise at timestep b on the DCT spectrum y_0 :

$$y_b = \sqrt{\alpha_b} y_0 + \sqrt{1 - \alpha_b} \epsilon$$

And we use the following equation as our loss function:

$$L = E_{x,b,\epsilon} [\|\epsilon - \epsilon_\theta(\sqrt{\alpha_b} DCT(x) + \sqrt{1 - \alpha_b} \epsilon, b)\|_2^2]$$



Visualization

The figure presents molecular dynamics trajectories generated for CC(=O)Oc1ccccc1C(=O)O, with an inset comparison to the reference conformations. The generated trajectories effectively capture the key conformational variations of the molecule over time and show good agreement with the reference structures. Overall, the generated conformations remain highly consistent with the ground truth, as evidenced by the low RMSD values.

(a) Part of Performance comparison of GraphMD and other methods on rMD17.

Molecule	Methods	$C_{time} \uparrow$	RMSF $\downarrow$	$D_{AG} \downarrow$	RMSD $\downarrow$
Aspirin	MD-ET	7.42	1.12	3.74	0.89
	HermNet	7.44	2.41	6.32	1.57
	Allegro	6.81	1.26	3.58	1.13
	TorchMD-NET	7.90	1.87	7.36	1.79
	GraphMD	9.69	0.57	1.84	0.35
Azobenzene	MD-ET	7.87	1.15	3.96	1.02
	HermNet	7.32	2.12	6.74	1.34
	Allegro	7.49	0.94	5.12	1.46
	TorchMD-NET	8.06	1.58	6.94	1.86
	GraphMD	9.76	0.63	2.03	0.41
Ethanol	MD-ET	7.49	0.96	4.25	0.80
	HermNet	7.39	1.76	6.93	1.67
	Allegro	7.34	1.13	4.41	1.21
	TorchMD-NET	8.15	1.92	9.21	1.64
	GraphMD	9.74	0.56	1.97	0.23

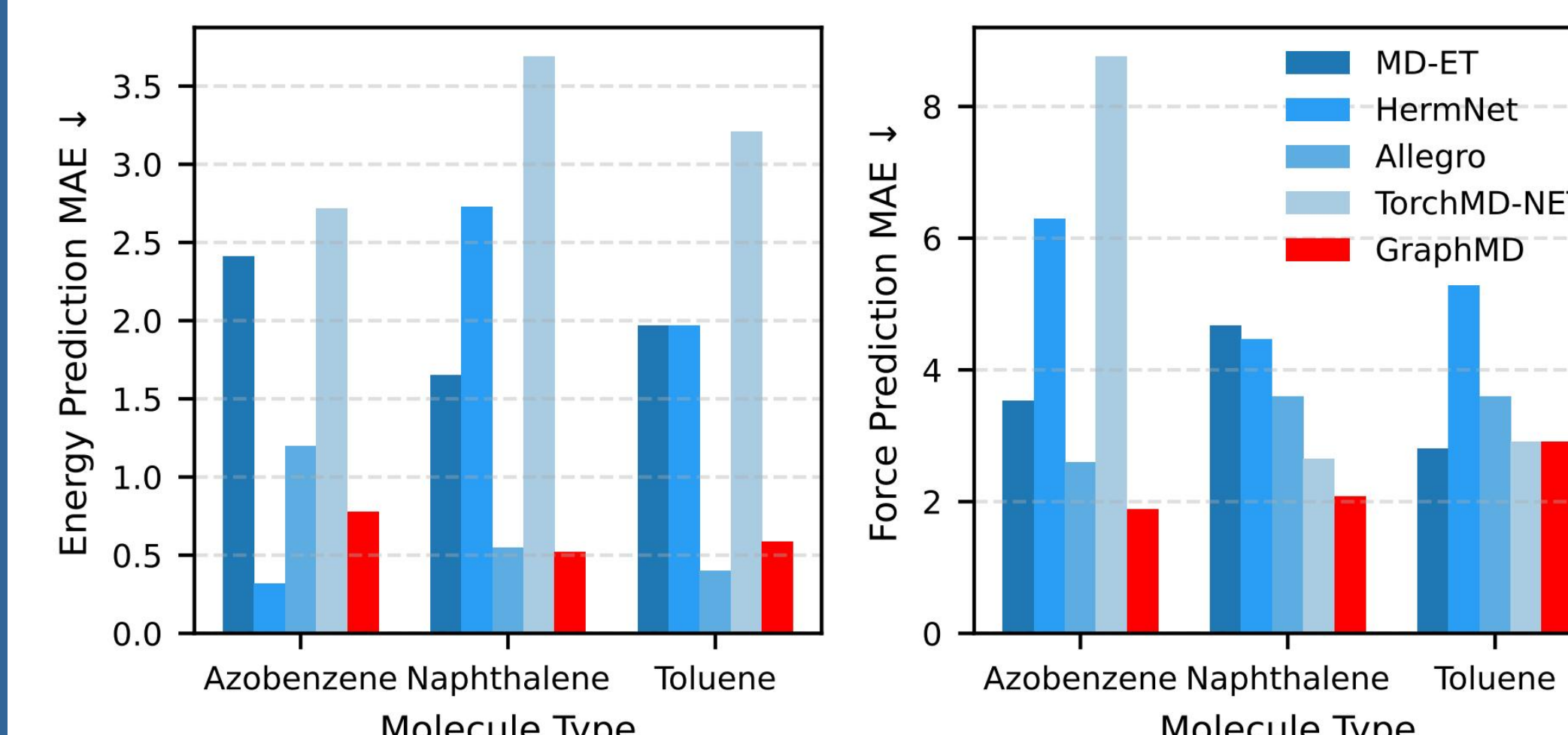
(b) Ablation study.

Molecule	Methods	$C_{time} \uparrow$	RMSF $\downarrow$	$D_{AG} \downarrow$	RMSD $\downarrow$
Azobenzene	Stage 1 + Stage 2	9.76	0.63	2.03	0.41
	Stage 1	9.57	1.78	2.27	0.89
	Stage 2	7.82	0.66	2.44	0.67
Naphthalene	Stage 1 + Stage 2	9.67	0.57	1.56	0.34
	Stage 1	9.62	1.43	1.99	0.70
	Stage 2	7.77	0.59	2.17	0.58
Toluene	Stage 1 + Stage 2	9.71	0.63	1.54	0.54
	Stage 1	9.70	1.58	2.63	0.69
	Stage 2	7.86	0.67	2.72	0.62

Results

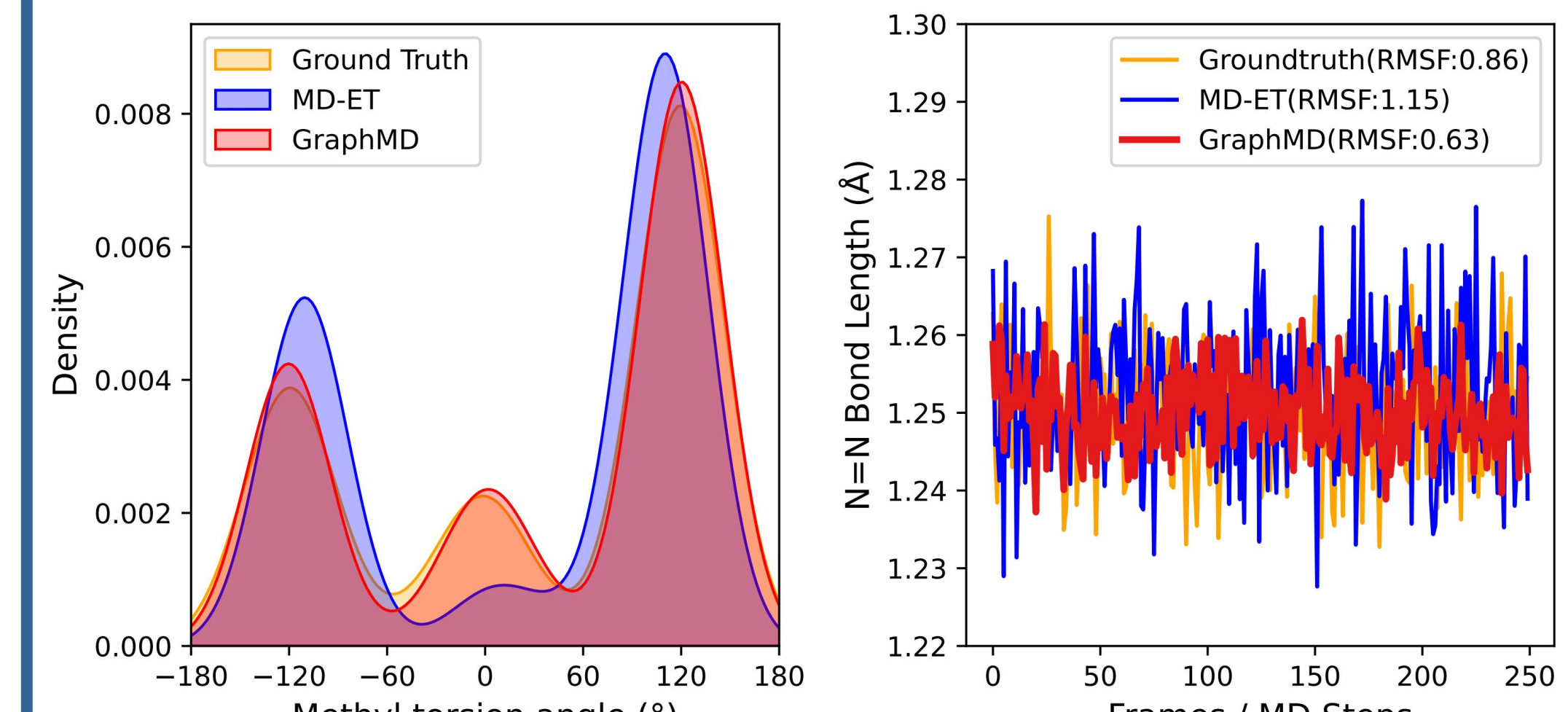
Table (a) compares GraphMD with existing methods under identical settings on the rMD17 datasets. GraphMD consistently outperforms SOTA baselines. Notably, for challenging cases such as ethanol and malondialdehyde, GraphMD surpasses the strong MD-ET baseline, improving C_{time} by 30.4% and RMSF by 41.7% (ethanol), and C_{time} by 33.2% and RMSF by 36.1% (malondialdehyde).

Table (b) shows ablation experiments on three randomly selected molecules to evaluate the contributions of the potential module (Stage 1) and DCT (Stage 2). Results show that combining both stages yields better performance than using either alone, confirming their effectiveness.



Validation 1

To further assess the effectiveness and reliability of our model in molecular modeling, we evaluated its performance on predicting the energies and forces of generated conformations. As shown in figure, GraphMD achieves state-of-the-art accuracy in force prediction and delivers competitive results in energy prediction compared to existing methods. These results demonstrate the strong capability and reliability of GraphMD in molecular property modeling.



Validation 2

To intuitively validate the effectiveness of our model in molecular structure learning and trajectory modeling, we conducted several visualization experiments. As shown in figure (left), the hydroxyl torsion angle distribution of ethanol generated by GraphMD closely matches the ground truth (GT) and outperforms the baseline MD-ET, benefiting from the DCT module's capacity to capture low-frequency torsional motions. For azobenzene, the temporal evolution of the N=N bond length.

Conclusion

In this study, we propose GraphMD, a diffusion-based deep learning framework for molecular dynamics. By integrating DCT-encoded temporal trajectories with physics-guided Morse and LJ constraints, GraphMD effectively captures smooth oscillations, maintains spatial consistency, and ensures the physical plausibility of trajectories. Extensive experiments demonstrate that GraphMD outperforms existing methods, providing an efficient and accurate solution for simulating complex molecular systems. Future work could focus on extending GraphMD to larger molecular systems and further enhancing its generalization capability.